

Project Title: Smart Hospital Assistant Robot for Autonomous Medicine Delivery

Purpose

Objective: To design and develop an autonomous hospital assistant robot that can securely deliver medicines from a medical storage station to designated patient beds using QR code verification, obstacle detection and automated medicine compartment locking.

Scope: The robot is intended for hospitals, clinics, nursing homes and healthcare facilities where routine medicine delivery is performed repeatedly. In the prototype environment, a medicine storage area and multiple patient beds will be simulated. The robot will autonomously travel to the medicine storage station, verify the required medicine using QR codes, receive the medicine from hospital staff, securely transport it to the selected patient bed, verify the patient's identity using another QR code and then release the medicine only to the correct patient.

Significance: Medicine delivery is a repetitive task that consumes valuable time for healthcare workers. This project demonstrates how robotics, embedded systems, computer vision and automation can be integrated to improve delivery efficiency, reduce human error, enhance medicine security and minimize unnecessary contact in healthcare environments. The project also serves as a practical implementation of autonomous robotic systems for real-world healthcare applications.

Components

Microcontroller:

- ESP32 Development Board
- ESP32-CAM Module

Sensors:

- IR Line Following Sensors
- HC-SR04 Ultrasonic Sensor
- ESP32-CAM Camera Module (QR Code Detection)

Actuators:

- DC Gear Motors
- SG90 Servo Motor (Medicine Compartment Lock)
- Active Buzzer
- LED

Body/Chassis:

- 2WD Robot Chassis Kit
- Lockable Medicine Storage Compartment

Additional Components:

- L298N Motor Driver
- 16×2 LCD Display with I2C Module
- Batteries
- Power Switch
- Breadboard
- Jumper Wires
- Resistors
- Buck Converter
- Acrylic/Cardboard Medicine Box
- PC for QR Code Processing and Task Management

Cost Breakdown (Approx.)

Component	Unit Cost (BDT)	Quantity	Total (BDT)
ESP32 DevKit (30pins)	700	2	700
ESP32-CAM	900	1	900
2WD Robot Chassis Kit	750	1	750
L298N Motor Driver	200	1	200
IR Line Following Sensors	60	3	180
HC-SR04 Ultrasonic Sensor	120	1	120
SG90 Servo Motor	150	3	150
16×2 LCD with I2C	350	1	350
Active Buzzer	50	1	50
LEDs and Resistors	48	1 Set	48
LiPo Battery Pack	1100	2	2100
Battery Charger	900	1	900
Breadboard	120	1	120
Jumper Wires	120	1 Pack	120
Medicine Compartment Materials	200	1	200
Mounting Hardware	100	1 Set	100

Estimated Total Cost: ≈ 6,000–7,000 BDT

Functionality Breakdown

Functionality 1: Autonomous Navigation and Obstacle Avoidance

Overview:

The robot autonomously travels between the medicine storage station and patient beds while continuously monitoring for obstacles.

Working Procedure:

The user selects the patient's bed number through a PC application. The PC sends the destination information to the ESP32. The robot follows a predefined hospital route using IR line-following sensors. The ESP32 continuously controls the DC motors through the L298N motor driver. During navigation, the ultrasonic sensor monitors the path ahead. If an obstacle is detected within the preset safety distance, the robot immediately stops and waits until the path becomes clear before resuming movement. Power is supplied through the rechargeable battery pack.

Functionality 2: Medicine Verification and Secure Loading

Overview:

The robot verifies the correct medicine request before beginning delivery.

Working Procedure:

The robot first travels to the medicine storage station. Each medicine box is associated with a QR code corresponding to a specific patient or prescription request. The ESP32-CAM captures the QR code and sends the image to the PC for processing and verification. Once the correct medicine is identified, hospital staff place the medicine inside the robot's lockable compartment. The ESP32 activates the servo motor to lock the compartment, ensuring that the medicine remains secure throughout transportation.

Functionality 3: Patient Verification and Medicine Delivery

Overview:

The robot verifies the patient's identity before releasing the medicine.

Working Procedure:

After receiving the medicine, the robot navigates to the selected patient bed. Upon arrival, the ESP32-CAM scans the QR code placed at the patient's bed. The captured image is processed by the PC and compared with the assigned destination information. If the QR code matches, the PC sends an authorization signal to the ESP32. The servo motor unlocks the medicine compartment, allowing the patient to retrieve the medicine. Simultaneously, the buzzer sounds, LEDs flash and the LCD displays the patient bed number and delivery confirmation message. If verification fails, the compartment remains locked and an error message is displayed.

Functionality 4: Return-to-Base Operation

Overview:

After successful medicine delivery, the robot automatically returns to the medicine station.

Working Procedure:

Once the medicine has been collected, the robot follows the predefined return route using the IR sensor array. Obstacle detection remains active throughout the return journey. Upon reaching the medicine station, the robot stops and enters standby mode. The LCD displays "Ready for Next Delivery", and the system waits for a new delivery request from the PC.